Automotive Technical Service Diagnostic Bulletin

Document ID: DIAG-S-CLASS-P0300
Classification: OEM Level 3 Diagnostic & Repair Procedure
Target Vehicle: Mercedes-Benz S-Class (W223 / S 500 4MATIC)
Powertrain: 3.0L Turbocharged Inline-6 (M256) with 48V Integrated Starter-Generator (ISG) & Electric Auxiliary Compressor (eZV)
Engine Control System: Bosch MG1CP002 (Engine Control Module N3/10) - FlexRay / CAN-FD Gateway
Electrical Architecture: Dual 12V / 48V Mild-Hybrid Architecture
Fuel Injection Architecture: 200-350 bar Direct Injection (Piezo HP Fuel Injectors, dual HP fuel pumps)
Firing Order: 1-5-3-6-2-4

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1. DIAGNOSTIC TROUBLE CODE (DTC) SPECIFICATION

- **DTC Code:** `P0300`
- **Standard Fault Description:** Random / Multiple Cylinder Misfire Detected
- **System Definition:** Engine control module detects crankshaft acceleration irregularities across multiple random cylinders exceeding the emissions/catalyst damage threshold.
- **MIL (Malfunction Indicator Lamp) Status:** Illuminated (Continuous or Flashing depending on catalyst severity)
- **Monitoring Condition:** Continuous monitoring during engine running cycle with coolant temp > 60°C and vehicle speed > 0 km/h.

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2. PLATFORM-SPECIFIC ROOT CAUSE ANALYSIS (RCA)

Root cause analysis customized specifically for the **Mercedes-Benz S-Class (W223 / S 500 4MATIC)**:

1. **48V DC/DC Ripple & Transient Noise:** Fluctuations on the 48V onboard electrical network passing through the DC/DC converter (N83/1) disrupting the primary ignition coil supply rail (Circuit 87M).
2. **Fuel Rail Pressure Imbalance across Dual Feed Rails:** High-pressure pump volume control valve (Y94) hunting under partial load, resulting in instantaneous lean misfire across multiple cylinders.
3. **Electric Auxiliary Compressor (eZV) Surge:** Momentary unmetered air puff through the bypass flap causing transient AFR swings prior to turbocharger spool-up.

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3. DEDICATED WIRING DIAGRAM & CIRCUIT PINOUT

Schematic configuration and electrical pin connections specifically for the S-Class control harness:

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[N3/10 ECM (Bosch MG1)]                [Ignition Coils T1/1 - T1/6]
Pin 12 (Cyl 1 Trigger) ----------> Pin 3 (Signal) [T1/1]
Pin 13 (Cyl 2 Trigger) ----------> Pin 3 (Signal) [T1/2]
Pin 14 (Cyl 3 Trigger) ----------> Pin 3 (Signal) [T1/3]
Circuit 87M (+12V Relay F32/3) --> Pin 1 (+12V Supply) [All Coils]
Ground W10/4 (Engine Ground)  --> Pin 2 (Power Ground) [All Coils]
Pin 28 (Ion Sense / Feedback) <-- Pin 4 (Combustion Monitor Bus)
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- Pinout Voltage Test: Pin 1 to Ground = 11.8V - 14.4V; Pin 3 to Ground = 0-5V square wave logic trigger.

Electrical Reference Points

- **Primary Ground Reference:** `W10/4 (Cylinder Head Ground, Left) & W16/5 (Front Bulkhead Ground)`
- **Power Distribution Node:** `Front SAM (Signal Acquisition Module N10/6) Fuse Box F32/3`

- **Signal Logic Voltage Range:** 0.0V to 5.0V regulated DC (unless high-voltage direct injection rail)

4. STEP-BY-STEP DIAGNOSTIC & VERIFICATION PROCEDURE

Follow this OEM workshop sequence to isolate the failure on this platform:

1. Connect XENTRY Diagnostic System; execute quick test and navigate to N3/10 -> Actuations -> Smooth Running Analysis.
2. Monitor rail pressure setpoint vs actual (B4/25 fuel pressure sensor); ensure steady ~250 bar at idle without >10 bar oscillations.
3. Verify 48V Battery (G1/3) state of health and DC/DC converter N83/1 filter capacitors for ripple voltage exceeding 150mV AC.
4. Perform software check on ME control unit N3/10; re-flash with latest flash calibration file for multispark ignition dwell map.

5. PRECISE RESOLUTION & REPAIR PROCEDURE

- Replace high-pressure fuel rail regulating valve Y94 if pressure fluctuation exceeds ±12 bar.
- Inspect ground stud W10/4 for galvanic oxidation; torque to 9 Nm.
- Reprogram N3/10 ME-SPEC control unit with updated ignition firing sequence adaptation parameters.

6. POST-REPAIR QUALITY VALIDATION & RE-TESTING

1. Clear all DTCs from Engine Control Module (`Bosch`) and secondary gateway modules.
2. Perform comprehensive drive cycle:
3. Cold start idle for 3 minutes (monitors secondary air / catalyst preheat and cold misfire).
4. Accelerate smoothly to 50 km/h and hold for 2 minutes.
5. Cruise at 80–100 km/h on highway for 10 minutes (validates downstream O2 sensor and fuel trim closed-loop balance).
6. Coasting deceleration (fuel cut-off check) for 5 seconds without braking.
7. Re-scan control units with diagnostic tool; confirm all OBD-II readiness monitors show **READY** / **COMPLETE** and no pending DTCs are stored.